RIG-CLOSED POINTS SURVIVE SUITABLE LOCALIZATION OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces the rig-closedness of the localized prime to a finiteness statement for the residue field, cites the finite-type residue field lemma, and applies the Nullstellensatz to choose f making the extension finite

Editorial note. The label lemma-rig-closed-point-after-localization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- rig-closed primes and the localization B_{f} of adic rings with prime q' over q (lemma-rig-closed-point-in-localization of the same chapter)
- the equivalence characterization of rig-closed primes via finite residue field extensions (lemma-rig-closed-point-relative of the same chapter)
- the residue field kappa of B/q is a finitely generated algebra over A/a (lemma-rig-closed-point-relative-residue-field of the same chapter)
- the Hilbert Nullstellensatz form: a field finitely generated over a domain becomes finite after inverting a suitable element (Algebra, Lemma algebra-lemma-field-finite-type-over-domain)

Source reference. Original lemma 00FY: sources/stacks-chow/tags.txt, source lines 588–588. Prerequisite pointer invoked at proof line(s) 3532; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGF: sources/stacks-restricted/pinned-original-source.tex, source lines 3460–3460. Prerequisite pointer invoked at proof line(s) 3521; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGB: sources/stacks-restricted/pinned-original-source.tex, source lines 3333–3333. Prerequisite pointer invoked at proof line(s) 3517; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGA: sources/stacks-restricted/pinned-original-source.tex, source lines 3314–3314. Prerequisite pointer invoked at proof line(s)

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3529; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\varphi : A \to B$ be an arrow of $WAdm^{Noeth}$ which is adic and topologically of finite type. Let $\mathfrak{q} \subset B$ be rig-closed. There exists an $f \in A$ which maps to a unit in $B/\mathfrak{q}$ such that we obtain a diagram*

$$\begin{array}{ccc} B & \longrightarrow & B_{\{f\}} \\ \uparrow{\scriptstyle \varphi} & & \uparrow{\scriptstyle \varphi_{\{f\}}} \\ A & \longrightarrow & A_{\{f\}} \end{array} \qquad \text{with primes} \qquad \begin{array}{ccc} \mathfrak{q} & \text{---} & \mathfrak{q}' \text{ ==== } \mathfrak{q}B_{\{f\}} \\ | & & | \\ \mathfrak{p} & \text{---} & \mathfrak{p}' \end{array}$$

such that $\mathfrak{p}'$ is rig-closed, i.e., the map $A_{\{f\}} \to B_{\{f\}}$ and the prime ideals $\mathfrak{q}'$ and $\mathfrak{p}'$ satisfy the equivalent conditions of Lemma 0GGB.

Proof. Please see Lemma 0GGF for the description of $\mathfrak{q}'$. The only assertion the lemma makes is that for a suitable choice of f the prime ideal $\mathfrak{p}'$ has the property $\dim((A_f)^\wedge/\mathfrak{p}') = 1$. By Lemma 0GGB this in turn just means that the residue field κ of $B/\mathfrak{q} = (B_f)^\wedge/\mathfrak{q}'$ is finite over $(A_f)^\wedge/\mathfrak{a}' = (A/\mathfrak{a})_f$. By Lemma 0GGA we know that $A/\mathfrak{a} \to \kappa$ is a finite type algebra homomorphism. By the Hilbert Nullstellensatz in the form of Algebra, Lemma 00FY we can find an $f \in A$ which maps to a unit in κ such that κ is finite over A_f. This finishes the proof. □